1、若 $A\in R^{n\times n}$ 满足 $A^T=A$，则称A为对称的

现有 $x\in R^n$，$f: R^n\to R$.

定义 $\nabla f(x)=\begin{bmatrix}\frac{\partial}{\partial x_1}f(x)\\ \vdots \\ \frac{\partial}{\partial x_n}f(x)\end{bmatrix}$，$\nabla^2 f(x)=\begin{bmatrix}\frac{\partial^2}{\partial x_1\partial x_1}f(x) & \cdots & \frac{\partial^2}{\partial x_1\partial x_n}f(x)\\ \vdots & \ddots & \vdots \\ \frac{\partial^2}{\partial x_n\partial x_1}f(x) & \cdots & \frac{\partial^2}{\partial x_n\partial x_n}f(x)\end{bmatrix}$

(a) 令 $f(x)=\frac{1}{2}x^TAx+b^Tx$，A是对称矩阵，b是n维向量，求 $\nabla f(x)$

$\nabla\frac{1}{2}x^TAx=\frac{1}{2}(A+A^T)x$，$\nabla b^Tx=b$

故 $\nabla f(x)=Ax+b$

(b) 令 $f(x)=g(h(x))$，$g: R\to R$ 且可微，$h: R^n\to R$ 且可微，求 $\nabla f(x)$

$\triangle f(x)=\nabla g(h(x))=\frac{\partial}{\partial x}g(h(x))=\left(\frac{\partial}{\partial h(x)}g(h(x))\right)\left(\frac{\partial}{\partial x}h(x)\right)=g'(h(x))\cdot\nabla h(x)$

(c) 令 $f(x)=\frac{1}{2}x^TAx+b^Tx$，A是对称矩阵，b是n维向量，求 $\nabla^2 f(x)$

$\nabla^2 f(x)=\nabla(\nabla f(x))=\nabla(Ax+b)=A$

(d) 令 $f(x)=g(a^Tx)$，$g: R\to R$ 为连续可微的，a为n维向量。求 $\nabla f(x)$，$\nabla g(x)$

$\nabla f(x)=\nabla g(a^Tx)=\frac{\partial}{\partial x}g(a^Tx)=\left(\frac{\partial}{\partial a^Tx}g(a^Tx)\right)\left(\frac{\partial}{\partial x}a^Tx\right)=g'(a^Tx)\cdot a^*$

$\cancel{\nabla^2 f(x)=\nabla(\nabla f(x))=\nabla g'(a^Tx)\cdot a^*}$

$\frac{\partial^2 g(h(x))}{\partial x_i\partial x_j}=\frac{\partial^2 g(h(x))}{\partial (h(x))^2}\cdot\frac{\partial h(x)}{\partial x_i}\cdot\frac{\partial h(x)}{\partial x_j}=g''(h(x))\cdot\frac{\partial h(x)}{\partial x_i}\frac{\partial h(x)}{\partial x_j}$

$\frac{\partial a^Tx}{\partial x_i}\cdot\frac{\partial a^Tx}{\partial x_j}=a_ia_j$，故 $\frac{\partial^2 g(a^Tx)}{\partial x_i\partial x_j}=g''(a^Tx)a_ia_j$

故 $\nabla^2 f(x)=\begin{bmatrix}a_{11} & \cdots & a_{1n}\\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn}\end{bmatrix}g''(a^Tx)=g''(a^Tx)\cdot a\cdot a^T$